Part 1

1. Rewrite this requirement so it is testable, and list two questions you'd ask the author: "Users should be able to reset their password easily."
    - A registered user should be able to start a password reset by entering their email address and the system should send them a single-use reset link to complete the reset by setting a new password that meets the password requirements.
    - Question 1: Should the user be asked to verify identity with anything, I.e. SMS or security question, or is just email reset link enough?
    - Question 2: Should there be a timeout on the process? For example, the reset link is only usable for 1 hour.
2. Classify each requirement as functional or non-functional:
    - (a) "The user can download an invoice as PDF."
        - Functional
    - (b) "The app works on Chrome, Firefox and Safari."
        - Non functional (compatibility)
    - (c) "The system sends an SMS when an order ships."
        - Functional
3. Label each as QA or QC:
    - (a) Running test cases on a build
        - QC
    - (b) Introducing a code review checklist
        - QA
    - (c) Training developers on secure coding
        - QA
    - (d) Inspecting a release candidate for defects
        - QC
4. Label each as verification or validation:
    - (a) Reviewing the design document
        - Verification
    - (b) A customer trying the app in user acceptance testing
        - Validation
    - (c) A peer reviewing code against the spec
        - Verification
5. Identify the error, defect, and failure: A developer thought VAT was 5% instead of 15%. The code calculates price * 1.05. Customers' invoices show too little tax.
    - ERROR: A developer thought VAT was 5% instead of 15%
    - DEFECT: The code calculates price * 1.05
    - FAILURE: Customers' invoices show too little tax

Part 2
Match each scenario to a testing principle:

- (a) "We ran the same 50 tests for a year and found nothing new." - 1 - Testing shows the presence of defects not the absence of bugs
- (b) "Most bugs came from the payment module." - 4 - Defects cluster together
- (c) "The app passed every test, but users found it useless." - 7 - Absence of errors fallacy

Manual or automated?

- (a) Checking a new logo looks right on mobile - Manual
- (b) Running 400 regression tests every night - Automated
- (c) Exploring a brand-new feature for the first time - Manual

Write **two positive** and **three negative** test cases for a "Sign up" form with Email and Password (password must be 8–16 characters).

Positive:

Email = adamfischer04@gmail.com
Password = Thisisapassword

Email = adamfishy7@gmail.com
Password = Thistoopassword

Negative:

Invalid email/Valid password
Email = NotanEmail
Password = Thiswouldwork

Valid email/invalid password
Email = Thiscouldbe@anemail.com
Password = NotWork

Invalid email/invalid password
Email = Thisisn't
Password = Nope

A movie ticket site gives prices by age: **0–12 = child, 13–59 = adult, 60+ = senior**. Identify the equivalence partitions (valid and invalid) and one test value for each.

Partitions:

Invalid: < 0
Valid:
- Child:
  - 0 <= age <= 12
- Adult:
  - 13 <= age <= 59
- Senior:
  - age > 59

Invalid test:
age = -2

Valid tests:
- Child:
  - age = 6
- Adult:
  - age = 22
- Senior:
  - age = 67

Part 3:

BVA: A text field accepts a username of **6 to 20 characters**. List the boundary values to test (three-value BVA) and the expected result for each.

Lower limit: 5,6,7
Upper limit: 19,20,21

Pass:
6,7
19,20

Fail:
5,21

Decision table: A bank approves a loan if the applicant has a salary over $30,000 **and** a good credit score. Applicants with only one of these are sent for manual review. Applicants with neither are rejected. Build the decision table.

	Case 1	Case 2	Case 3	Case 4
C1. Salary > 30,000	True	True	False	False
C2. Credit Score = Good	True	False	True	False
Approved	True			
Manual Review		True	True	
Rejected				True

State transition: A document can go **Draft** → **Submitted** → **Approved**, or **Submitted** → **Rejected** → **Draft** (edited and resubmitted). List the valid transitions and one invalid transition to test.
Valid:
Draft - submitted - approved
Draft - submitted - rejected - draft - approved
Draft - submitted - rejected - draft - rejected - draft - approved

Invalid:
Draft - Rejected - submitted - approved

Use case: For "User withdraws cash from an ATM", write the main flow (4–6 steps) and two exception flows.

Main Flow:
User inputs card
User types correct pin
User selects amount for withdrawal
User waits for cash to come out
User takes cash and card

Exception Flows:
User inputs card
User types incorrect pin
User retypes incorrect pin
Users card is blocked

User inputs card
User types correct pin
User selects amount for withdrawal above their current balance
User is informed they do not have that amount of current cash available

Error guessing: List five error-guessing tests for a "Phone number" field

1. Not a number
2. Not the correct area code
3. Too small number
4. Spaces or other characters in number
5. Too large a number